

Expanded Search for Toroidal Isomers

Bryan M. Harvey

Preliminary Exam / Master's Defense

2026-07-08

Liquid Drop Model (1970s) - Original Theory

Semi-Empirical Mass Formula

$$E_B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} \pm \delta(N, Z)$$

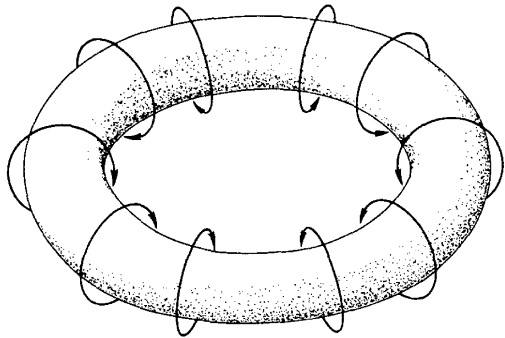
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- E_B depends on **spherical** assumptions.

What happens under deformed toroidal configurations at high angular momentum?



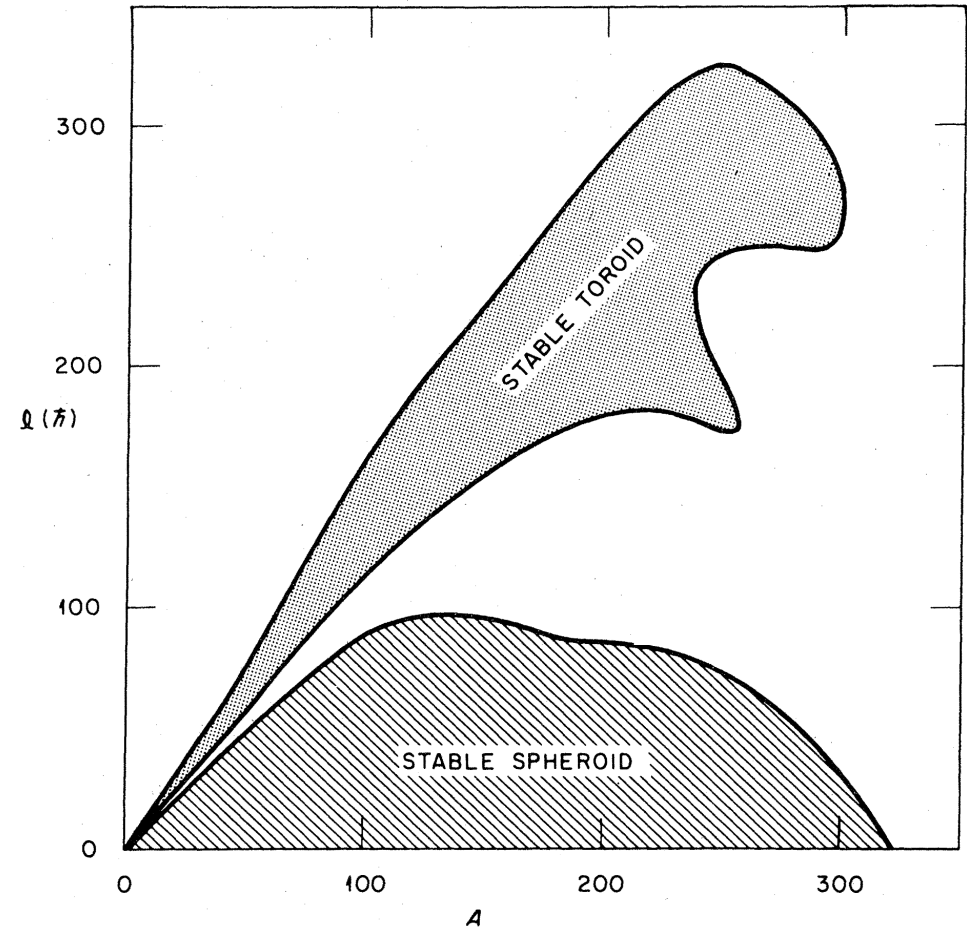
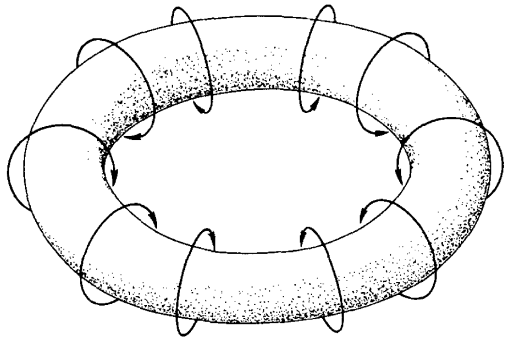
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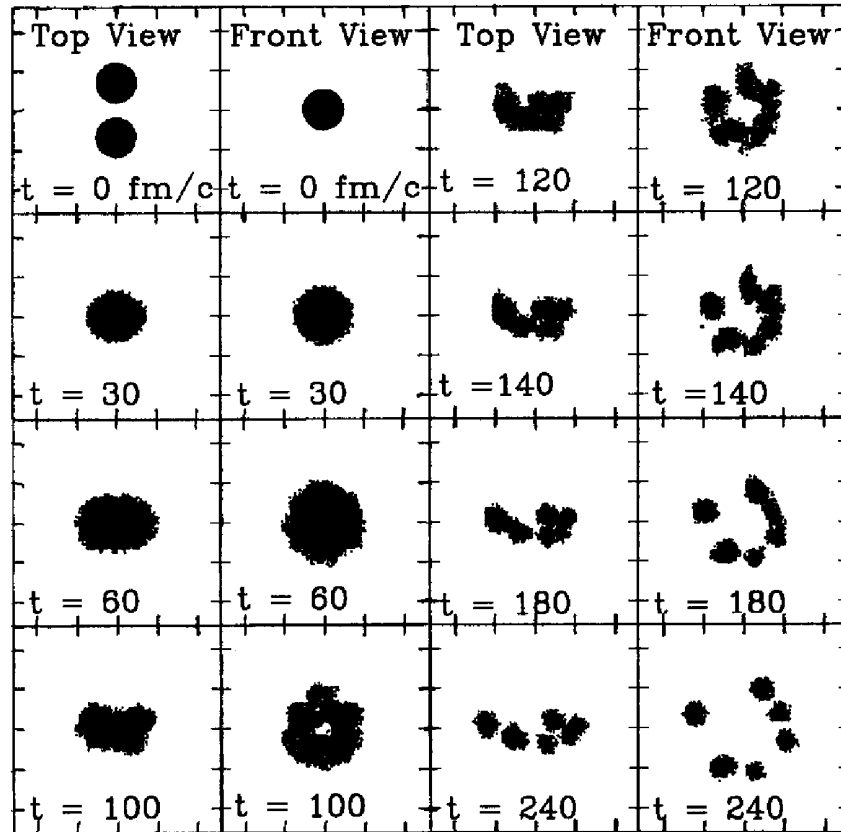
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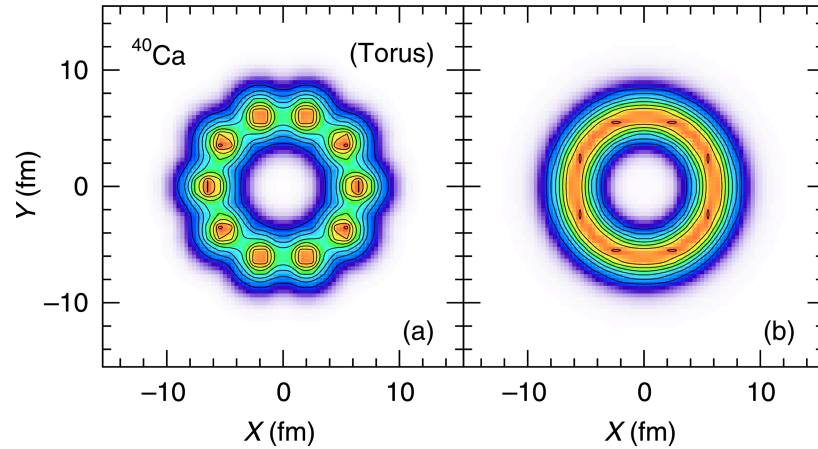
Boltzmann-Uehling-Uhlenbeck (BUU) (1990s) - Dynamic Formation

$^{92}\text{Mo} + ^{92}\text{Mo}$, $E/A = 100$ MeV, $b=0$, Stiff EOS



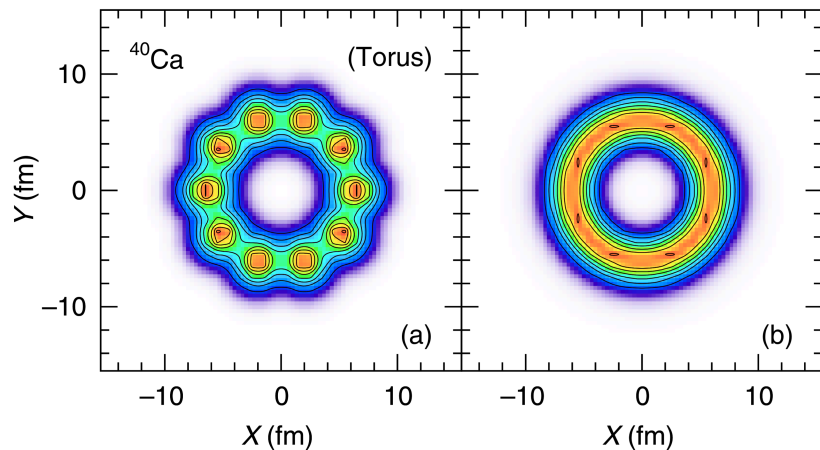
- Direct collisions produced outwardly expanding rings.
- Ring expansions temporarily slowed
- Eventual decay into symmetric particles
 - $r_{\text{particle}} \approx d_{\text{toroid}}$

Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific State Predictions



- Predicted state in ^{40}Ca
 - $J_z = 60\hbar$
 - $E^* \approx 175 \text{ MeV}$

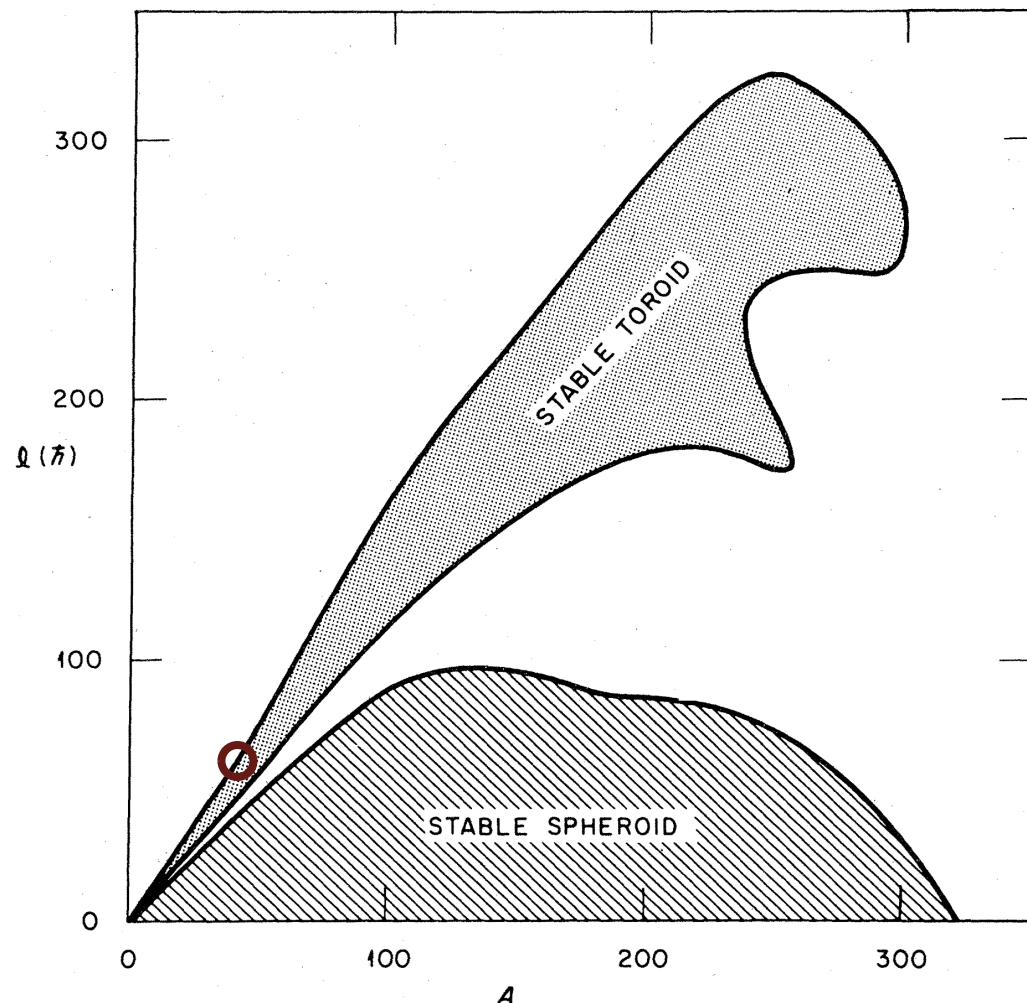
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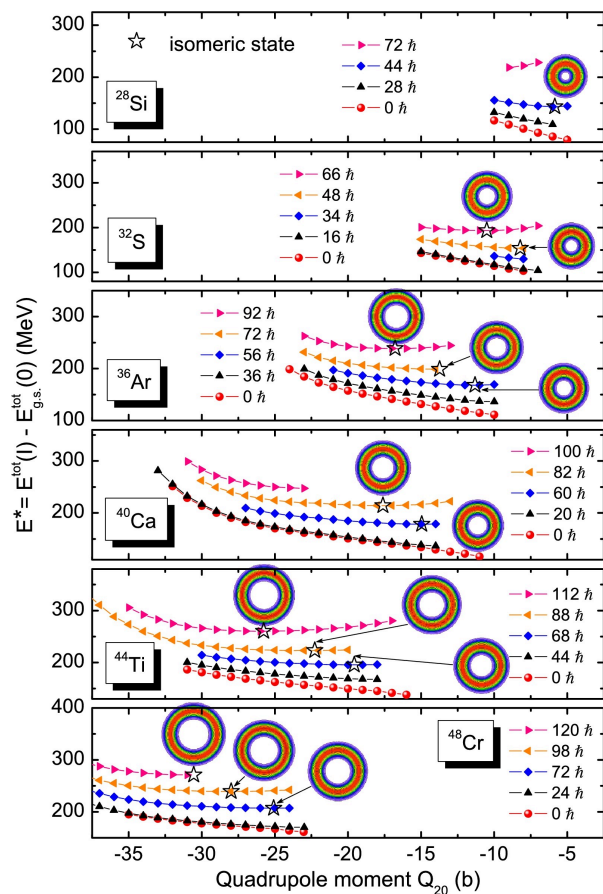
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Angular momentum and mass consistent
with Wong's LDM



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A. Staszczak and C.-Y. Wong predict 18 toroidal isomers

	E^* [MeV]	I [$\hbar$]	d [fm]	ρ_{max}/ρ_0
^{28}Si	143.18	44	1.45	0.74
^{32}S	153.87	48	1.42	0.76
	193.35	66	1.40	0.67
^{36}Ar	168.03	56	1.40	0.78
	198.63	72	1.39	0.71
	238.56	92	1.37	0.64
^{40}Ca	178.36	60	1.40	0.79
	214.23	82	1.39	0.73
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Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific State Predictions

- Ichikawa's state is reproduced
- Many predictions $\Rightarrow$ trends
 - Low-density states
 - Minor axis radius $\sim \alpha$ radius

Toroidal states, should they exist, might decay through several α particles

- Many states to look for

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Minor axis stays about constant ($d \approx 1.4$ fm)

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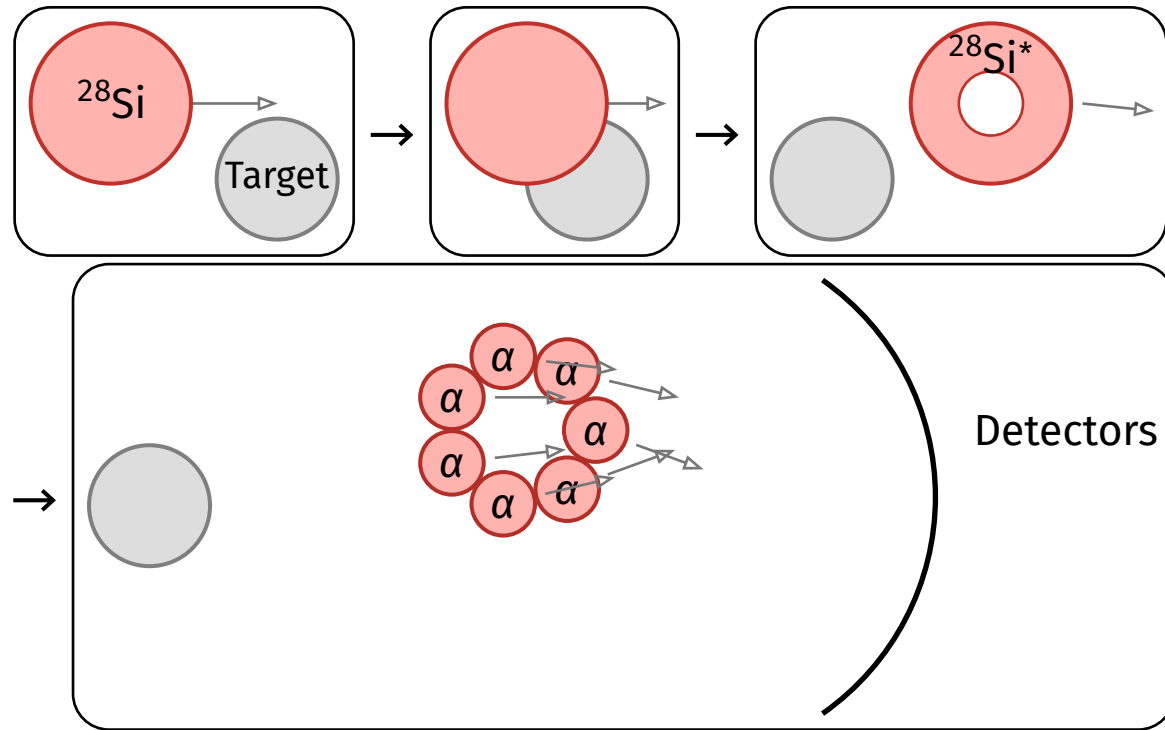
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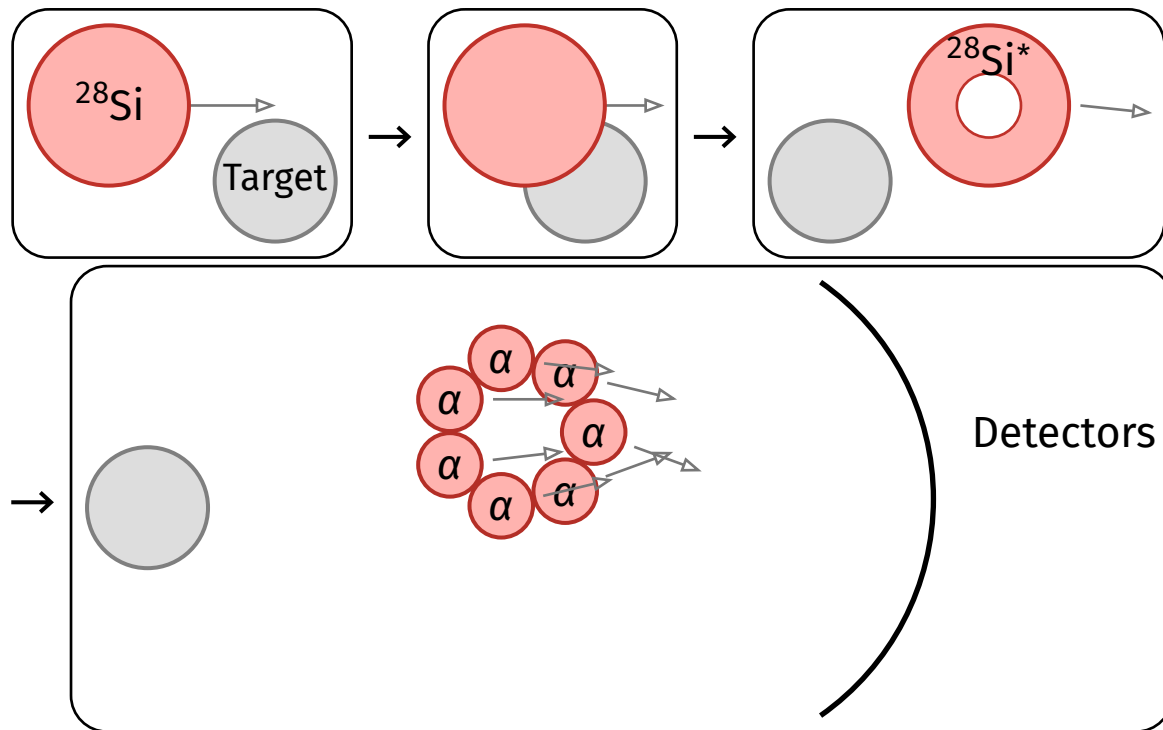
Several states to investigate, including $^{28}\text{Si}(E^* = 143 \text{ MeV})$

First Experimental Evidence (2019)



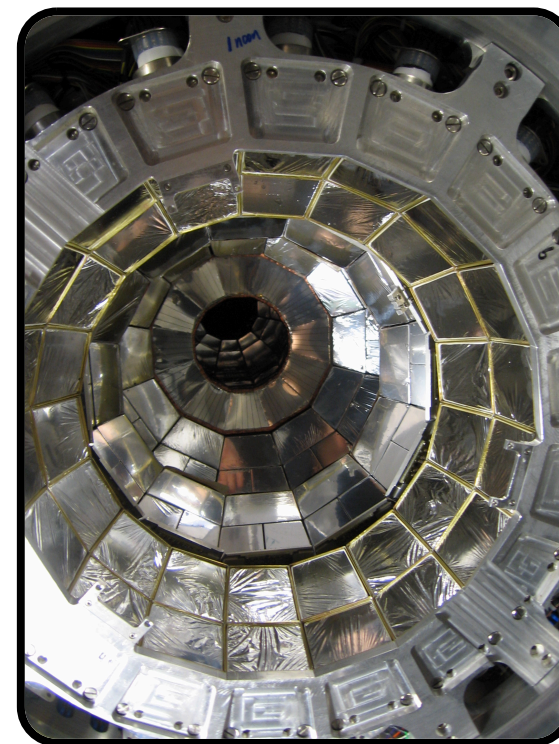
$$E_{^{28}\text{Si}}^* = \underbrace{\sum_i T_{\alpha_i}}_{E_{\text{rel}}} - \underbrace{(m_{^{28}\text{Si}} - 7m_{\alpha})}_Q \quad \text{if } 7\alpha \text{ from } ^{28}\text{Si}^*$$

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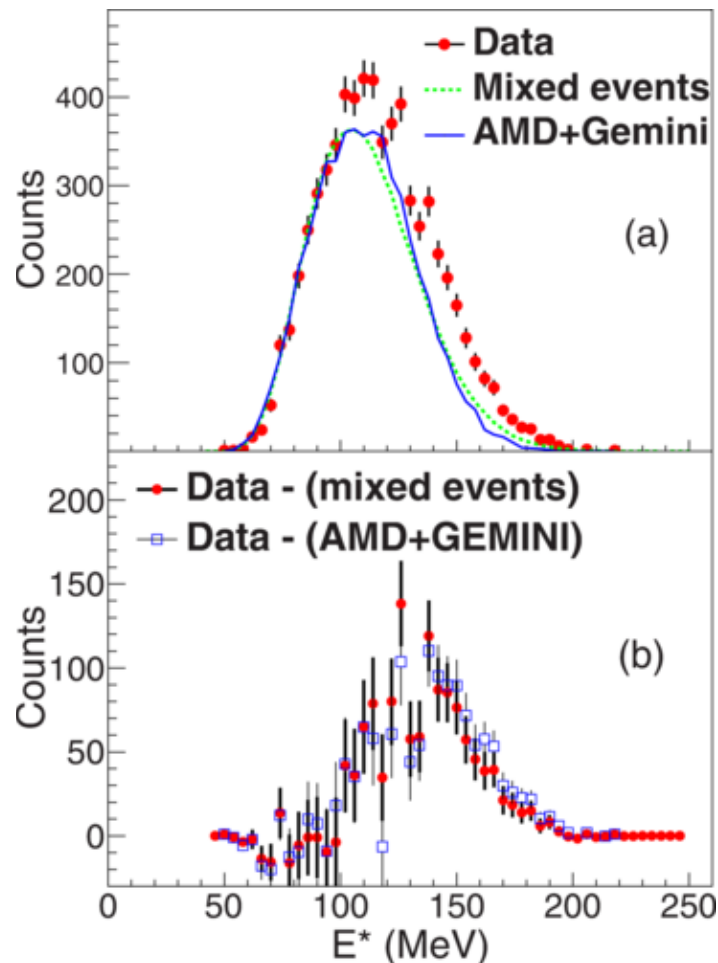
Neutron Ion Multidetector for Reaction Oriented Dynamics (NIMROD)



$^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u

First Experimental Evidence (2019) - Cao et al., JBN Group

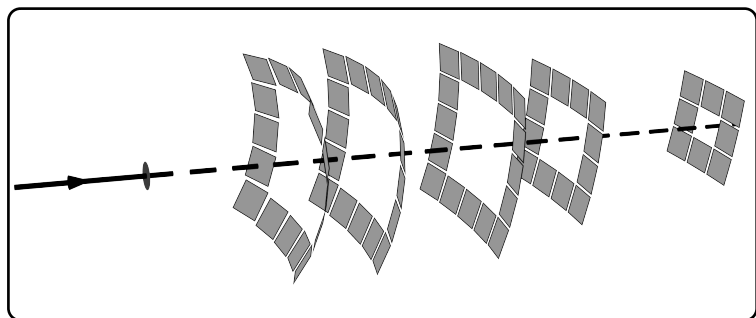
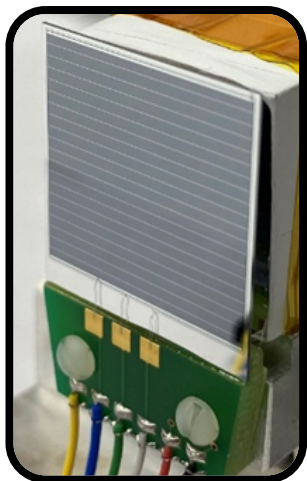
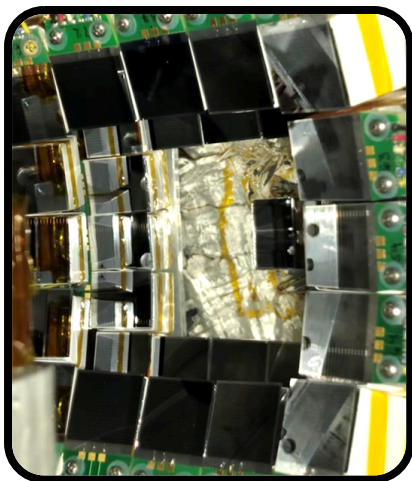
- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Part of 2009 experimental series
- Sample Size
 - 6,467 events w/ 7α
- Resolution for E^* from 7α
 - ~ 9.4 MeV (FWHM) in ROI
 - Position-insensitive detectors



- Nonresonant Background
 - Mixed Events
 - Shifted AMD Distribution
- Extracted Peaks
 - $E^* = 114, 126, \text{ \& } 138$ MeV
 - FWHM dominated by resolution

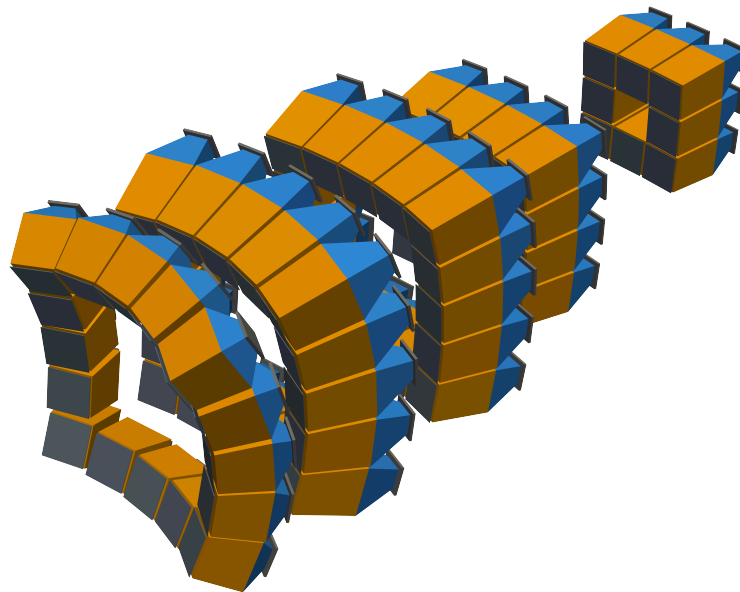
“Clearly an experiment with much better angular resolution, allowing better resolution for the excitation energy spectrum, will be very desirable.”

Forward Array Using Silicon Technology (FAUST)

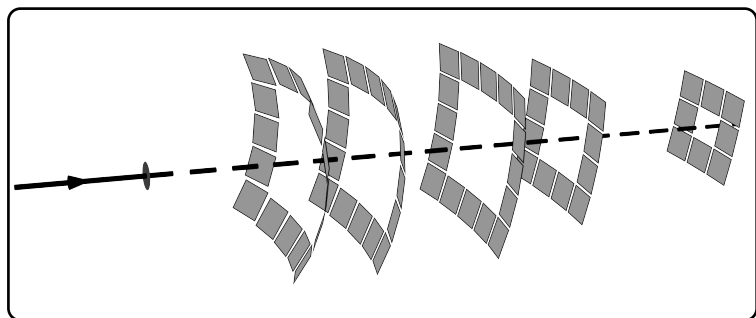
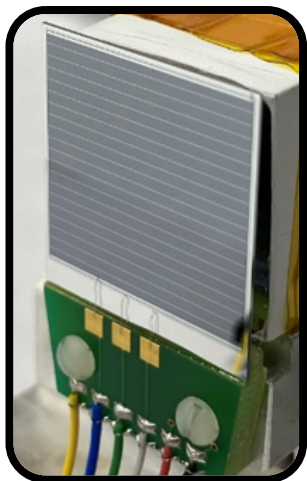
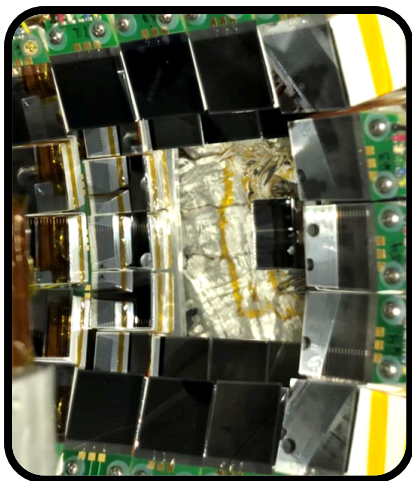


Forward Array Using Silicon Technology (FAUST)

- 68 Si-CsI(Tl) telescopes
 - Excellent forward coverage ($1.7^\circ - \sim 40^\circ$)
 - Lab energy (Thick CsI stops particles)
 - Isotopic particle identification ($E-\Delta E$ technique)
 - Lab angle (Dual-Axis Duo-Lateral detectors)



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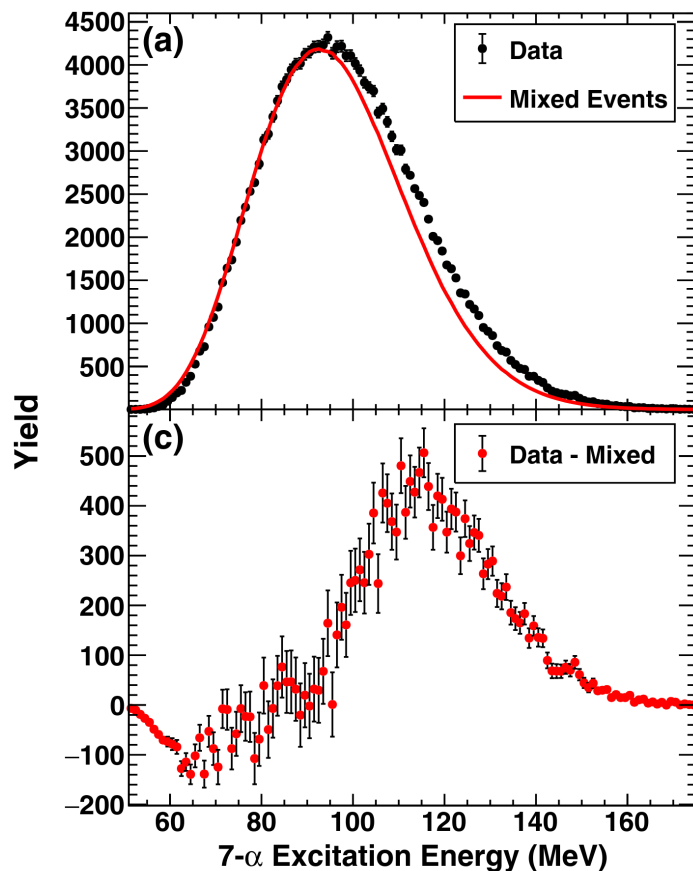
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Measured Reactions

- Target: ^{12}C
- Beam Energy: 35 MeV/u
- Projectiles
 - ^{16}O , ^{20}Ne , ^{24}Mg (Develop analytical techniques + future cluster studies)
 - ^{28}Si (Andy Hannaman's thesis work – confirm and characterize states from Cao *et al*)
 - ^{32}S , ^{36}Ar (Search for new states predicted by Wong)

Second Experimental Result (2024)

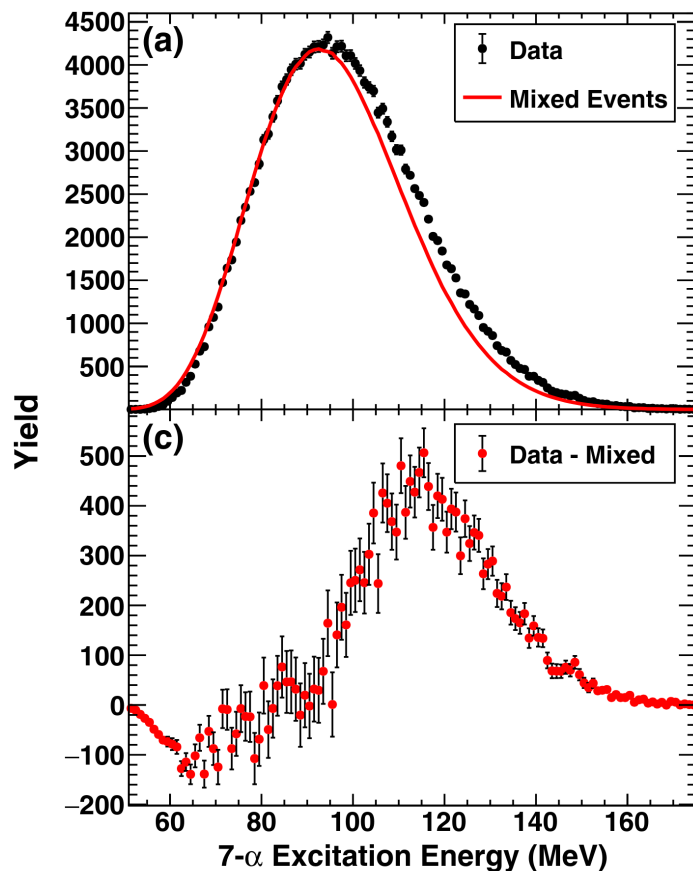
- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Dedicated measurement for toroidal search
- Sample Size
 - 187,067 Events w/ 7α (x29 $\uparrow$)
- Resolution for E^* from 7α
 - ~ 2.5 MeV (FWHM) in ROI (x3.5 $\downarrow$)



- **No obvious peaks in the raw spectrum.**
 - Despite being more sensitive measurement

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- **No obvious peaks in the raw spectrum.**
 - Despite being more sensitive measurement
- Simulated Background
 - Systematics hard to constrain
- Mixed Events Background
 - Known bias (see next slides)
- Original analysis suggests peaks must be smaller or broader than originally suggested.

A physically-motivated, validated, and data-driven background estimate is required for confident interpretation of the $N\alpha\ E^*$ spectra.

One way we know mixed events are imperfect (for this purpose)

A Case Study on d - d - d

Any signals should be **rare to nonexistent**

High-energy excitation, de-exciting by unlikely channel twice

- $Q(^6\text{Li} \rightarrow 3d) = -25.32 \text{ MeV}$
- $Q(\alpha \rightarrow 2d) = -23.84 \text{ MeV}$
- $Q(d \rightarrow pn) = -2.22 \text{ MeV}$

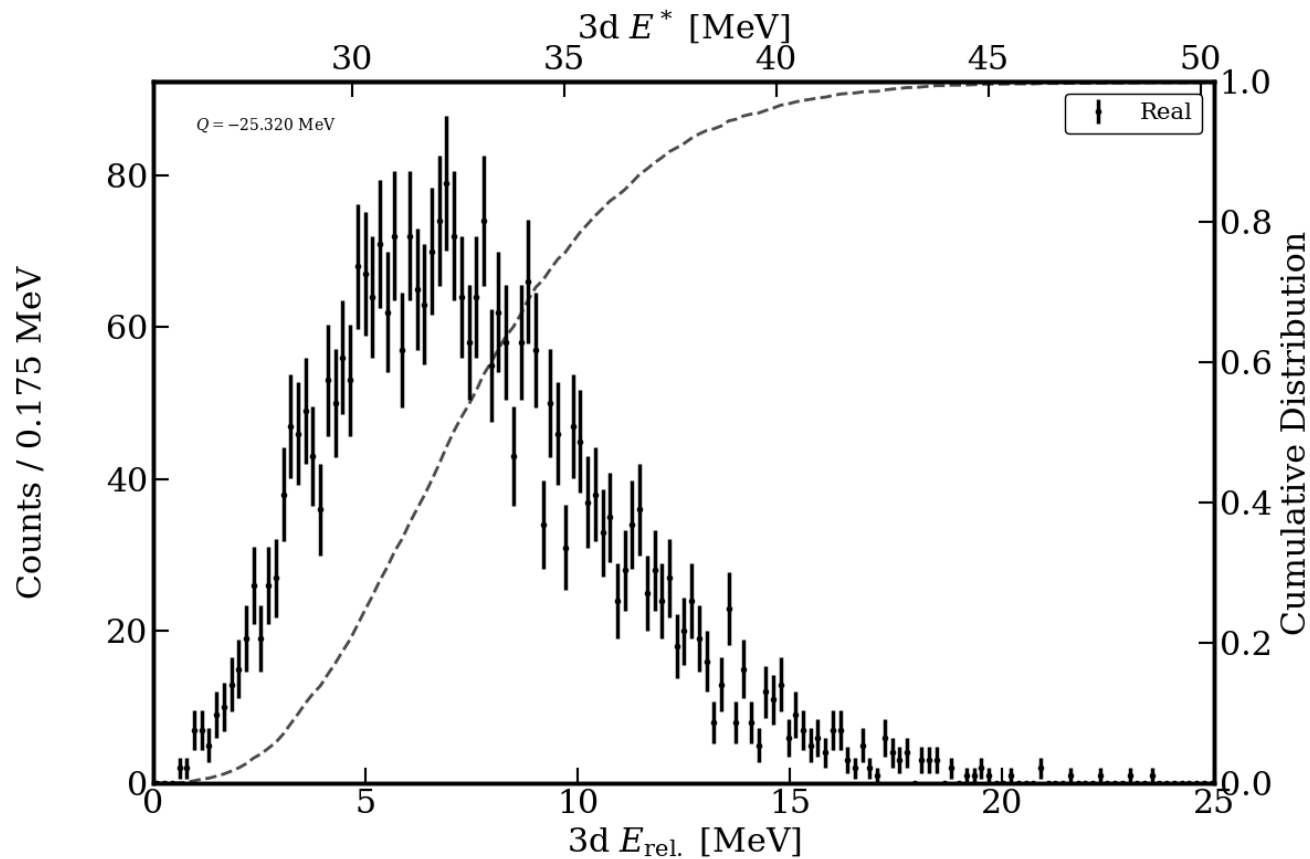
$^6\text{Li}(E^* > 25.32 \text{ MeV})$

$\rightarrow \alpha^*(E^* > 23.8 \text{ MeV})$

+ $d(E^* < 2.2 \text{ MeV})$

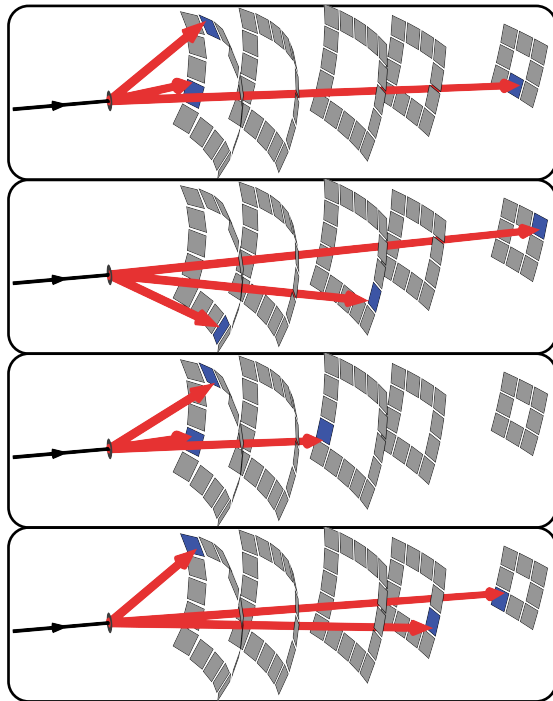
$\rightarrow 3d(E^* < 2.2 \text{ MeV})$

Total = Background + Signal



Creating a Mixed Event

Real Events

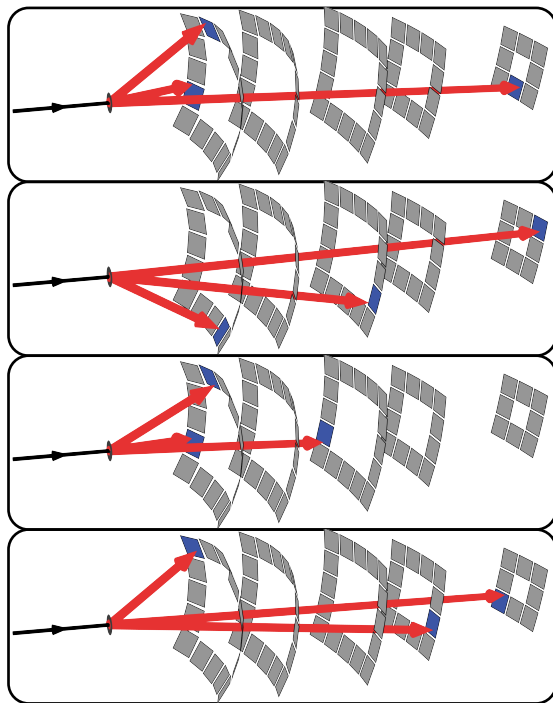


⋮

- Start w/ events containing all particles

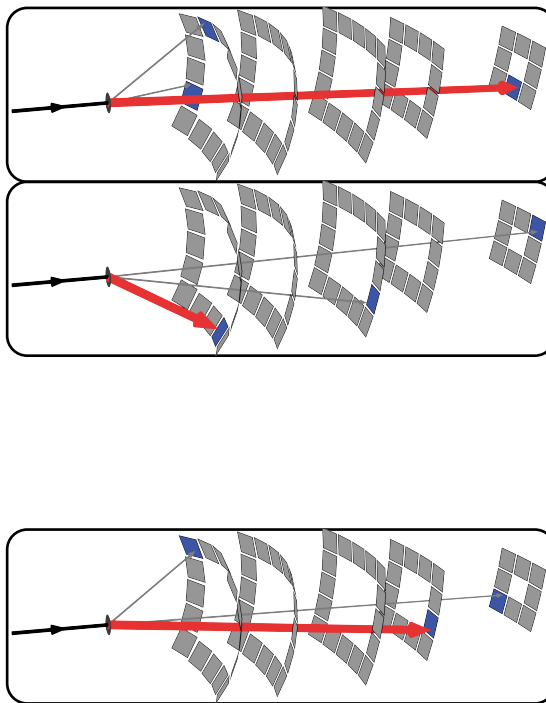
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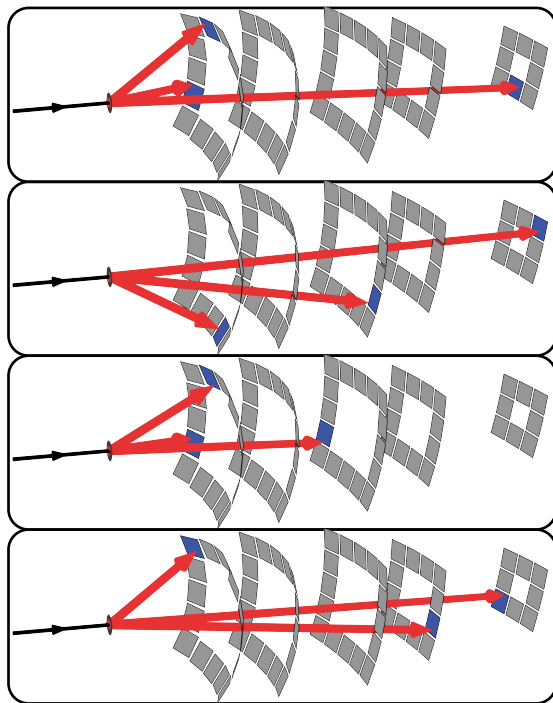
Selected Events



- Start w/ events containing all particles
- Select particles from separate events

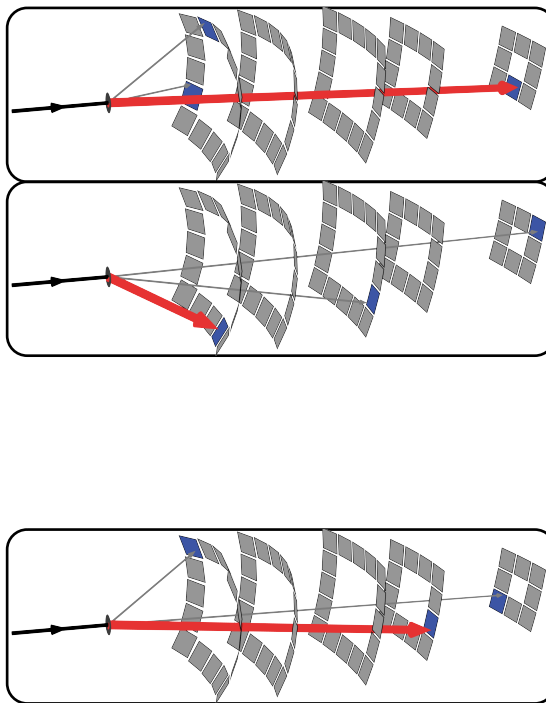
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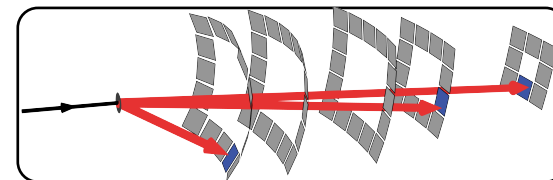


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Selected Events



Mixed Events



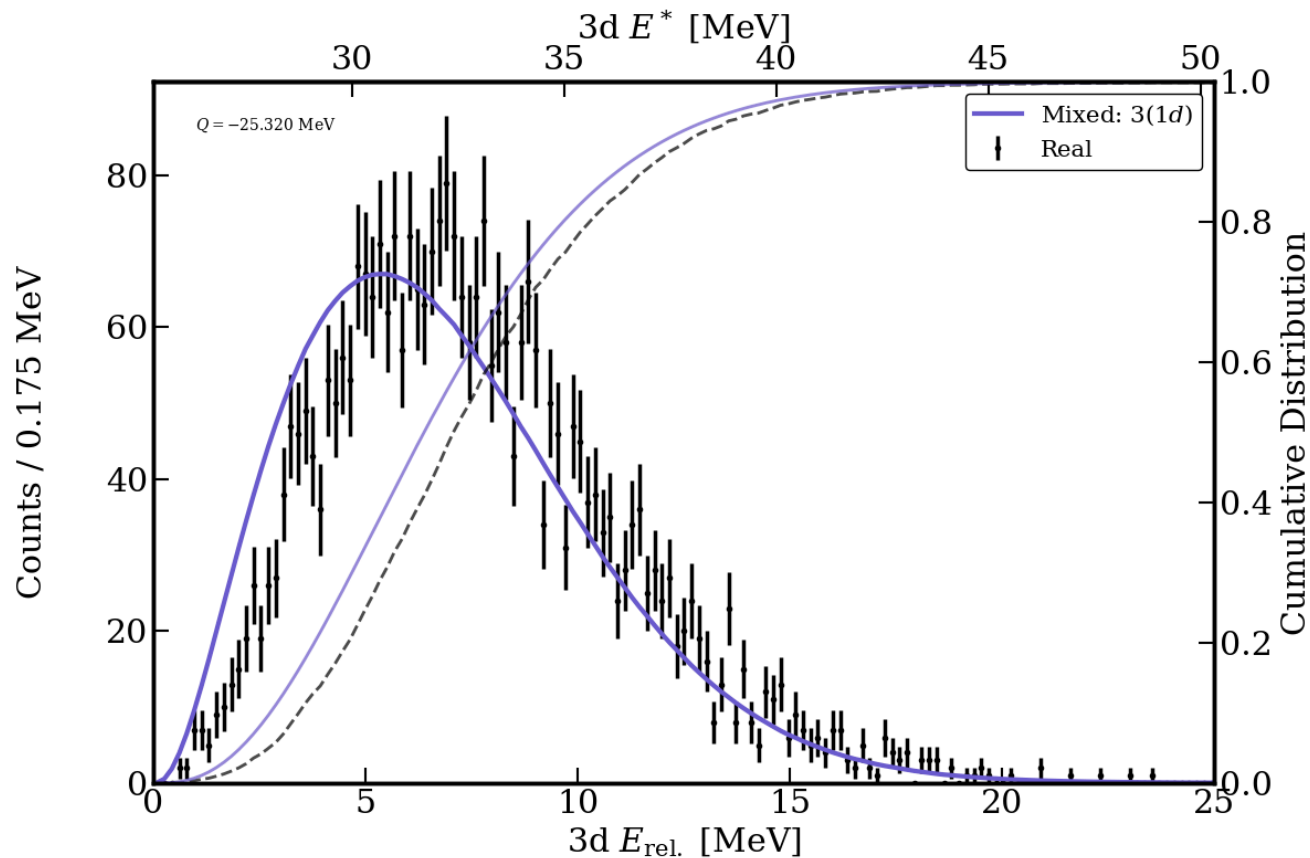
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- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

One way we know mixed events are imperfect (for this purpose)

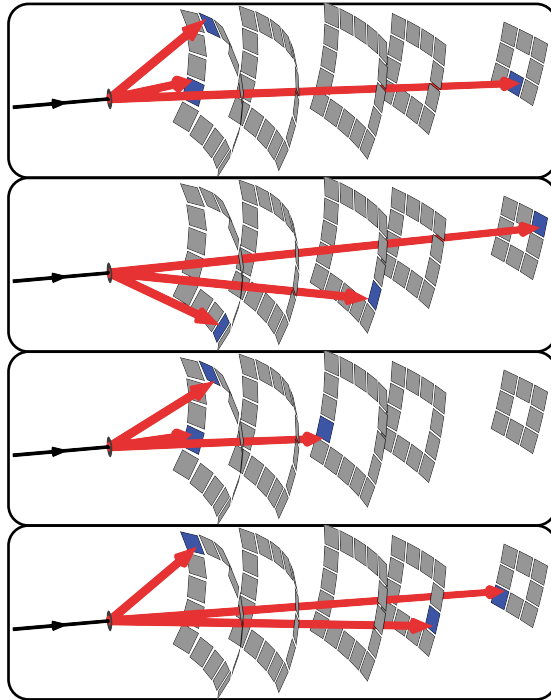
Mixed $\neq$ Total

- Mixing events removes particle-particle correlations, including:
 - 3-particle resonances (good)
 - Coulomb repulsion (bad)
 - 2-particle resonances (bad)
- Need a way to **remove 3-particle** correlations while **preserving 2-particle** correlations



Creating a *Partially Mixed Event*

Real Events

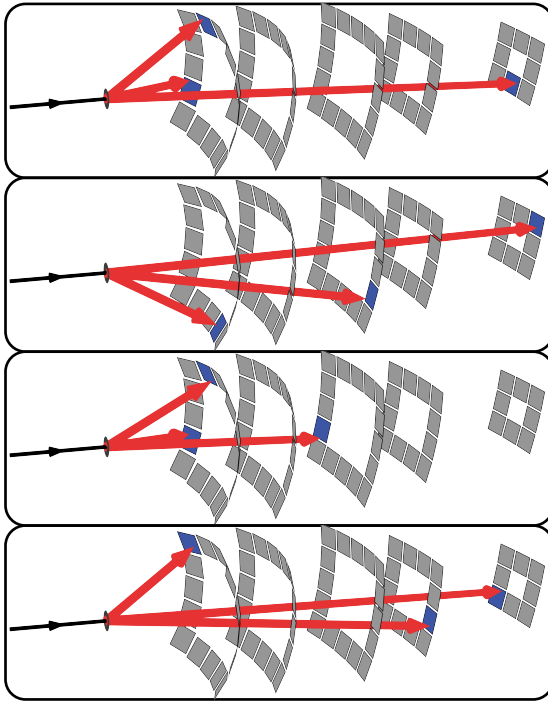


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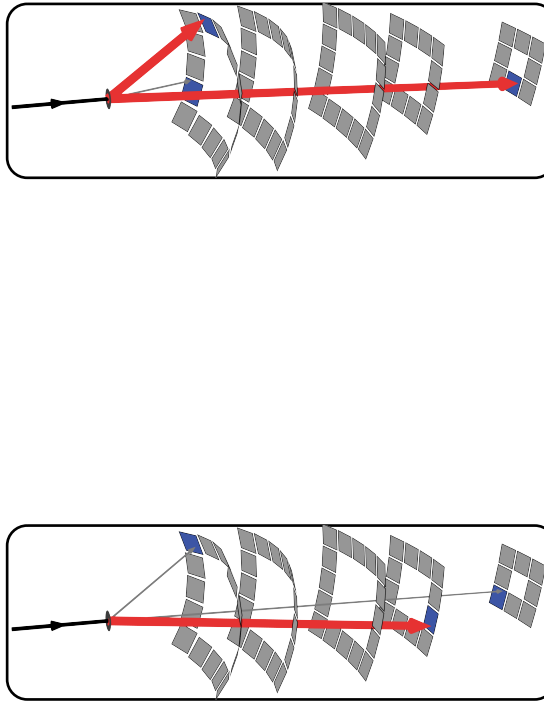
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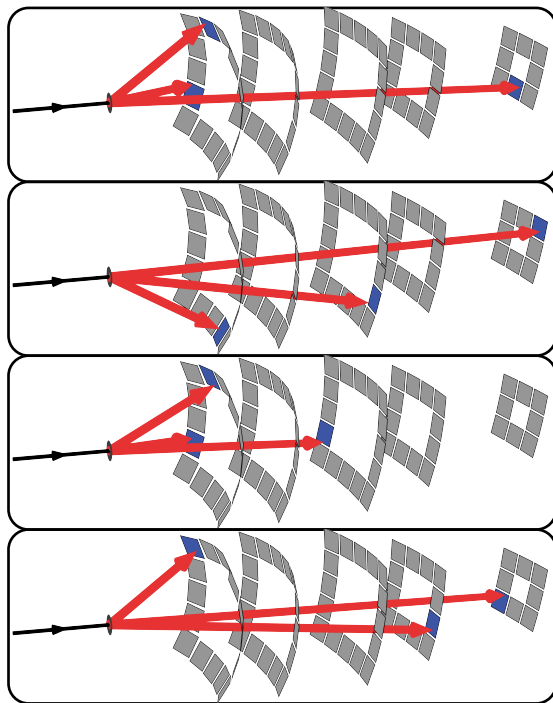
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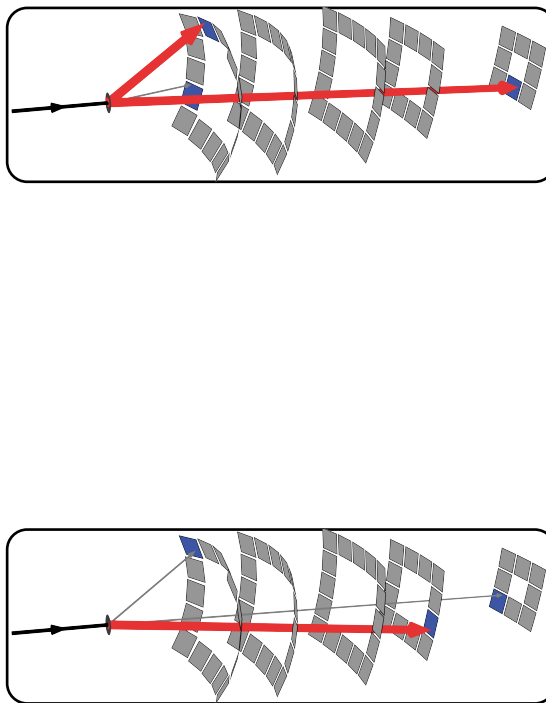
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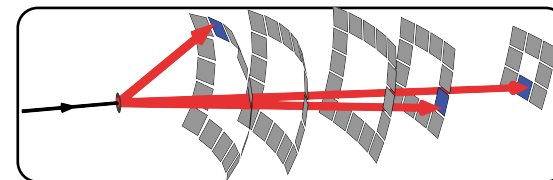


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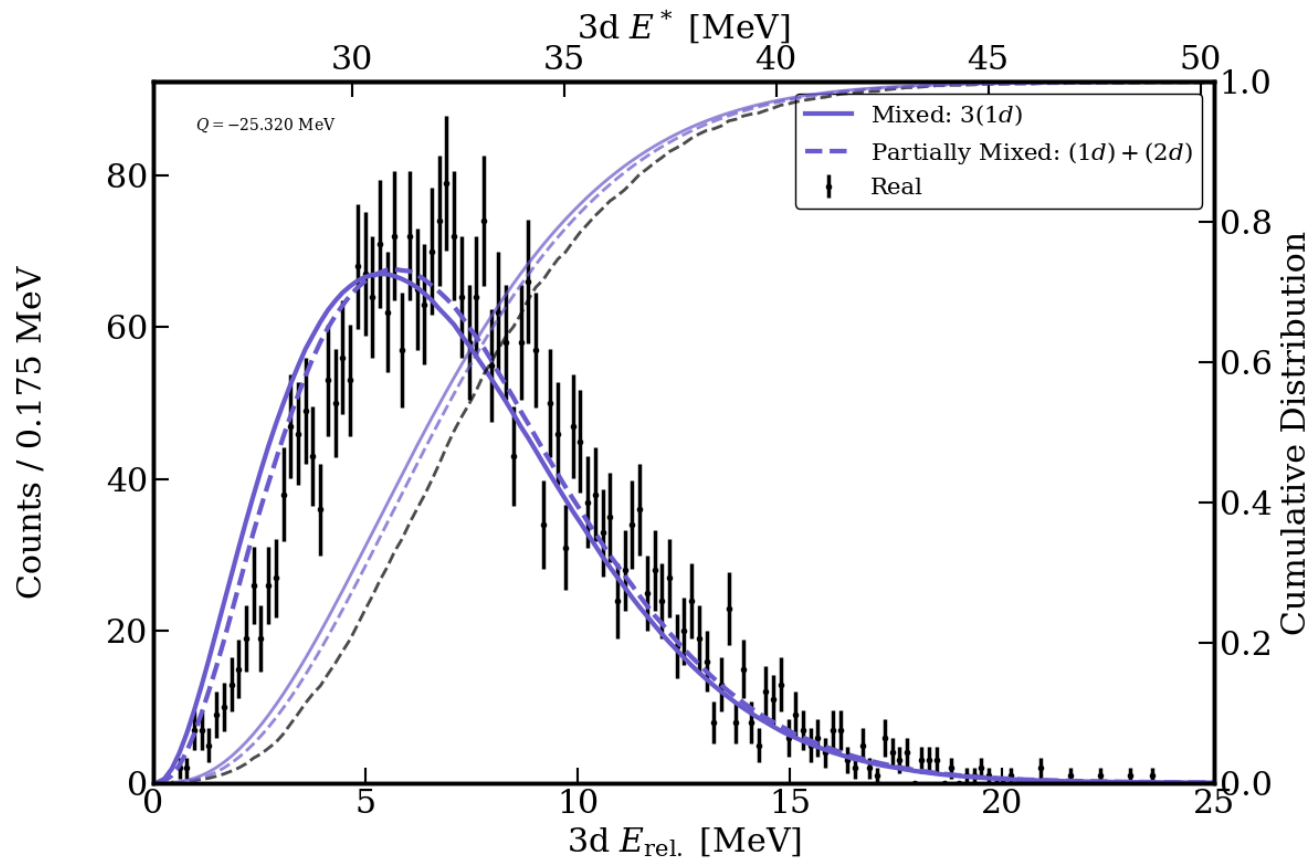


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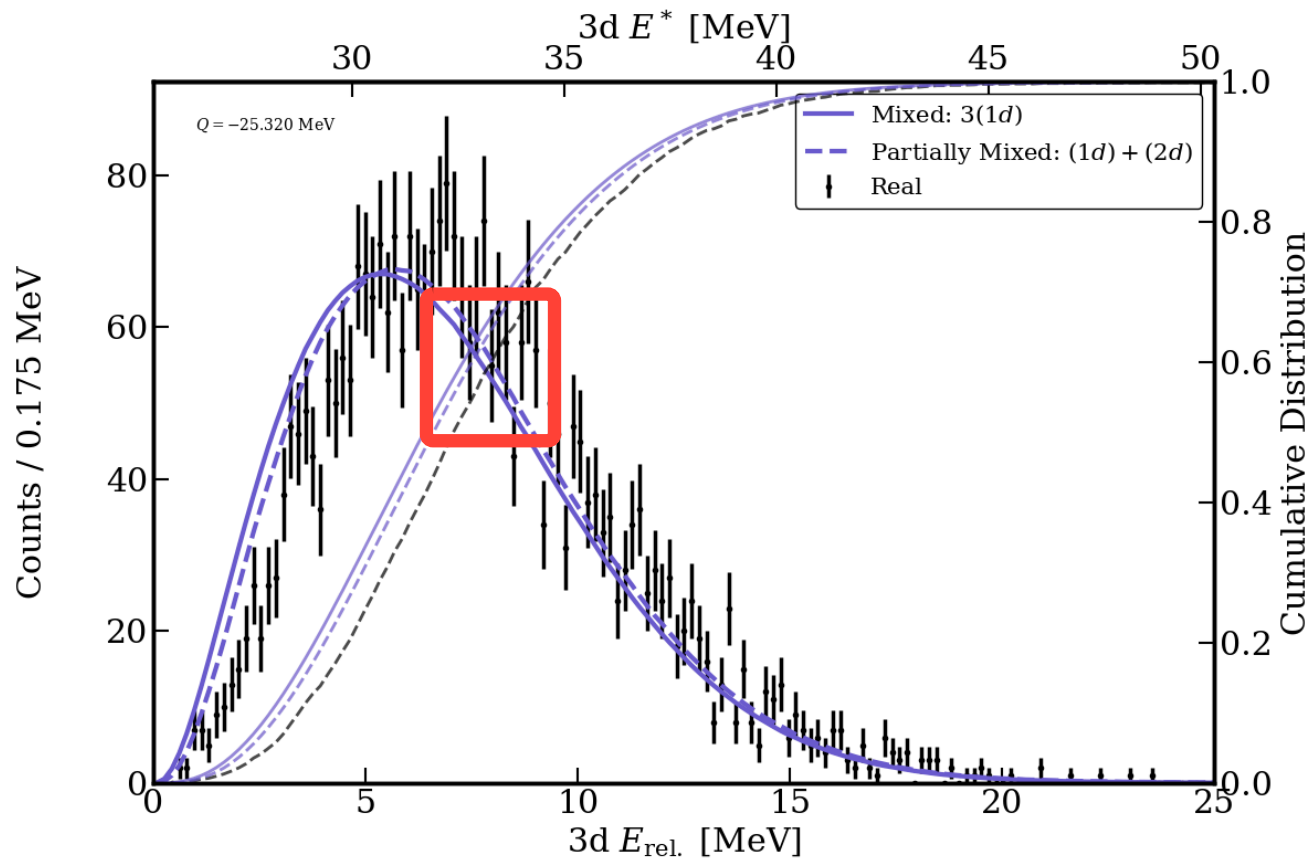
Incorporating one of the d - d correlations

- Partial mixing (PM) provides a minor improvement over full mixing (FM).
- PM is $\sim \frac{1}{3}$ of the way between FM and Real in the inverse CDF



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Measure the impact of one $d-d$ correlation

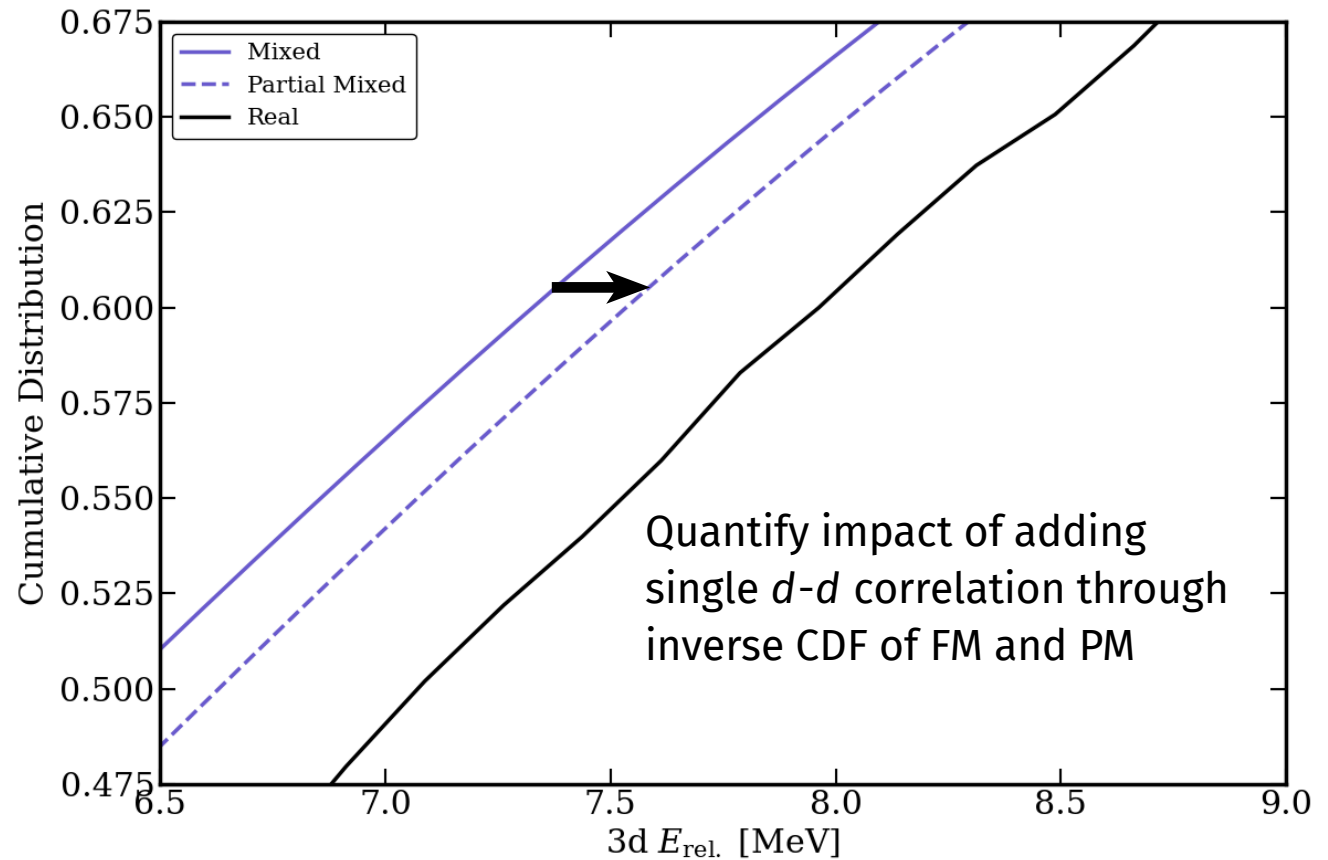
Number of $d-d$ correlations

Real: $\binom{3}{2} = 3$

Partially Mixed: $\binom{2}{2} + \binom{1}{2} = 1$

Mixed: $3\binom{1}{2} = 0$

- Quantile mapping: Horizontal shifts in the CDF
 - Frequently used in climate change studies



Extrapolate the impact to three $d-d$ correlations

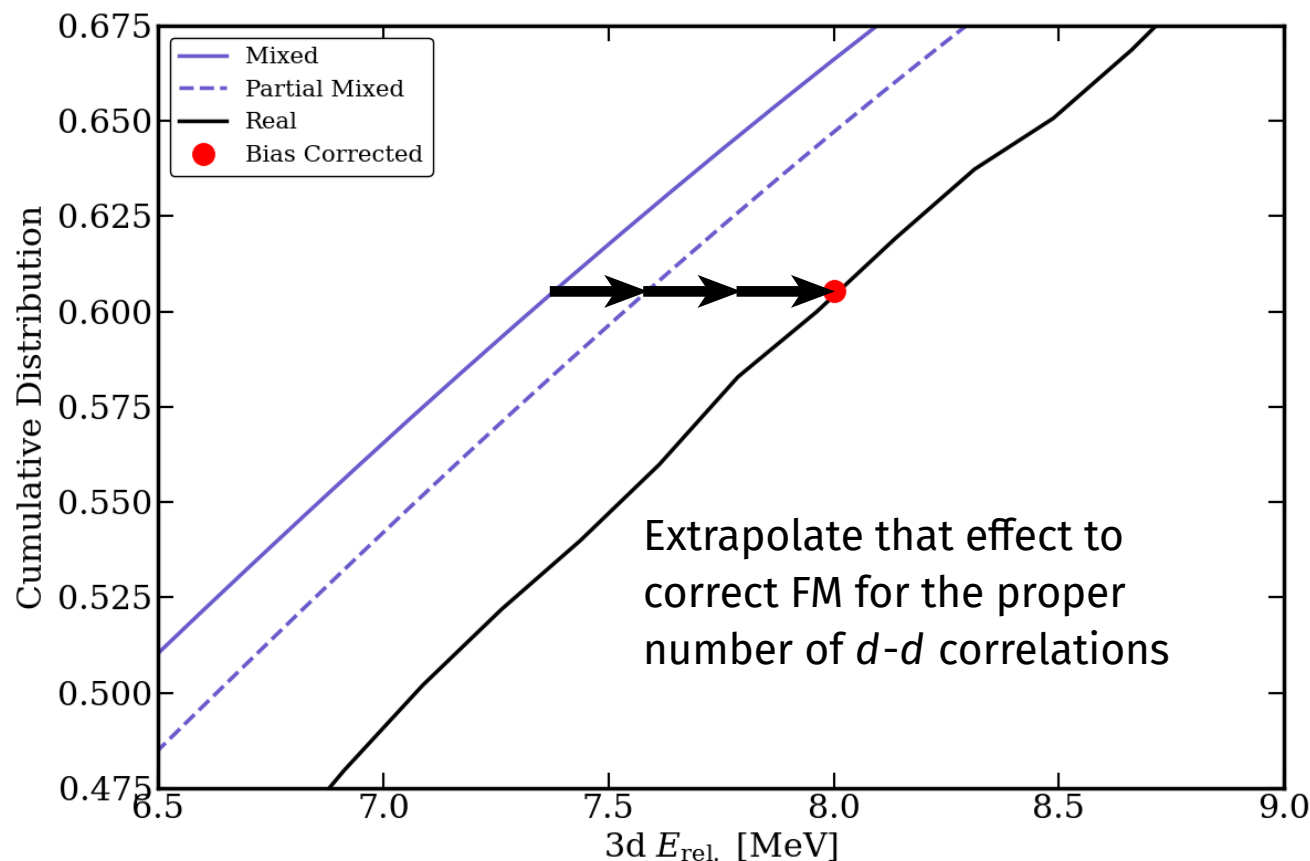
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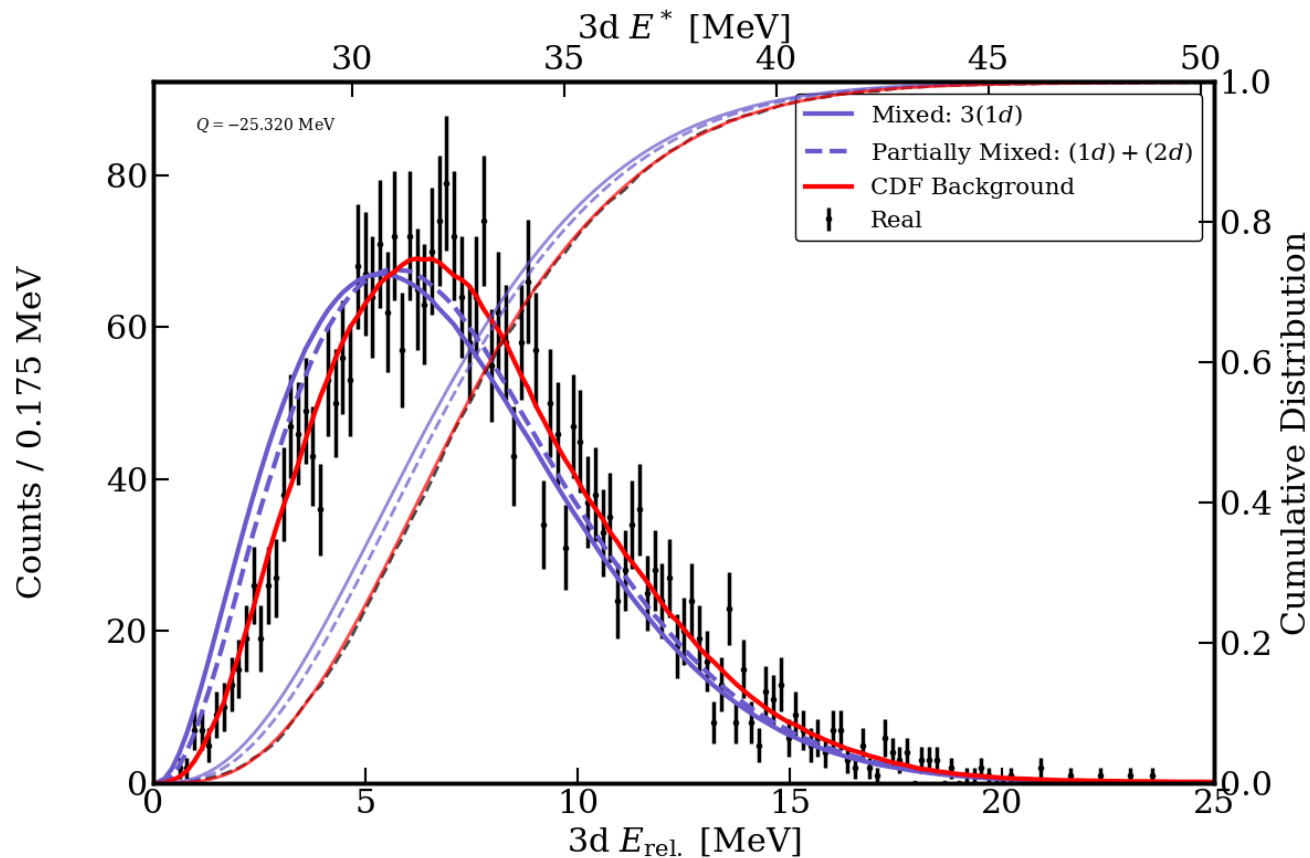
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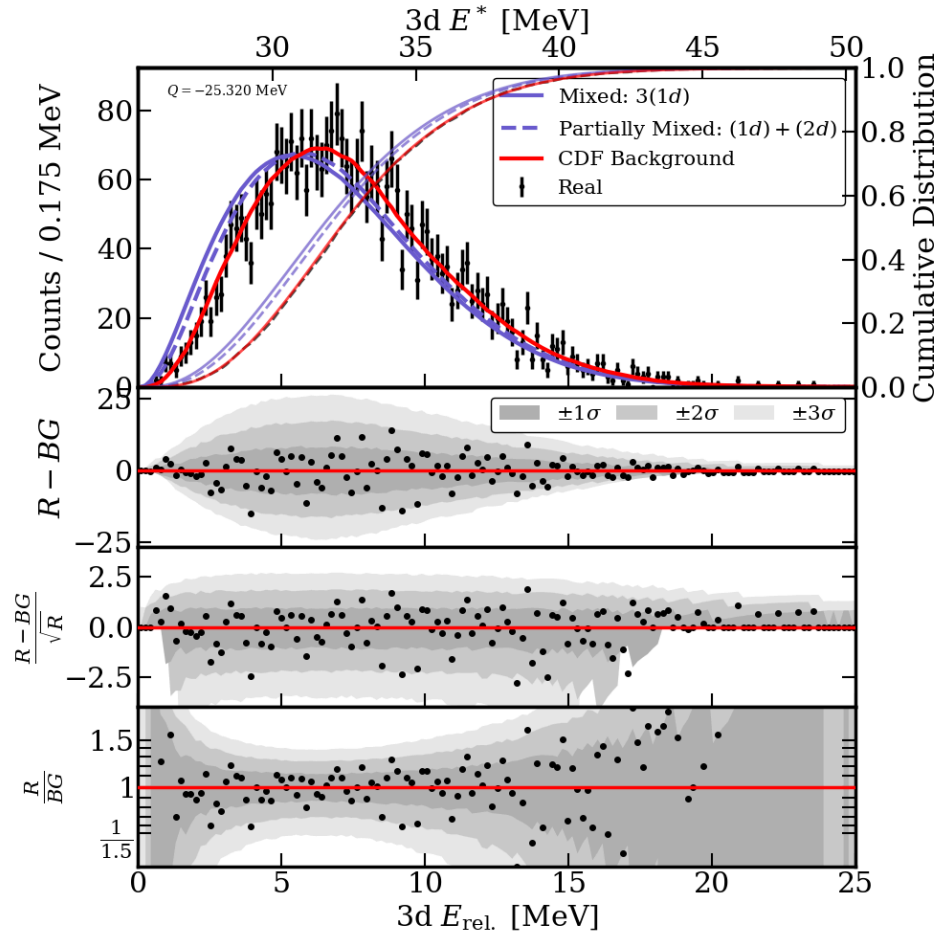
Derive new background through CDF

Novel Estimation of the Nonresonant Background

1. Measure the systematic impact of introducing a single 2-particle correlation, via the transformation from FM to PM
2. Propagate that systematic change an additional 2 times to account for all three 2-particle correlations



Statistical Check

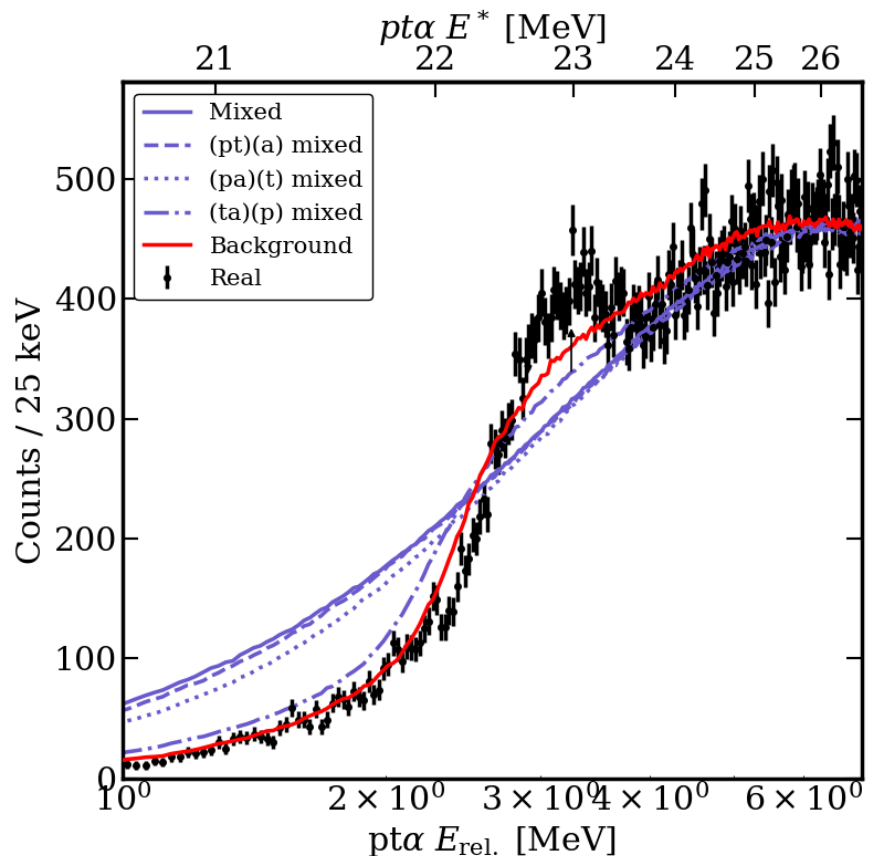


Does the **Background** match the **Total Real**?

- ‘Zero Parameter Fit’
- $\chi^2 / \text{dof} = 0.96$ (1 is ideal)
- $P = 0.4998$ (0.5 is ideal)
 - Resample the model and assess how often a more extreme χ^2 / dof is observed
- Residuals and ratios show no systematic issues

The background spectrum is statistically consistent with the measured spectrum.

Another Case Study: ${}^8\text{Be} \rightarrow p + t + \alpha$

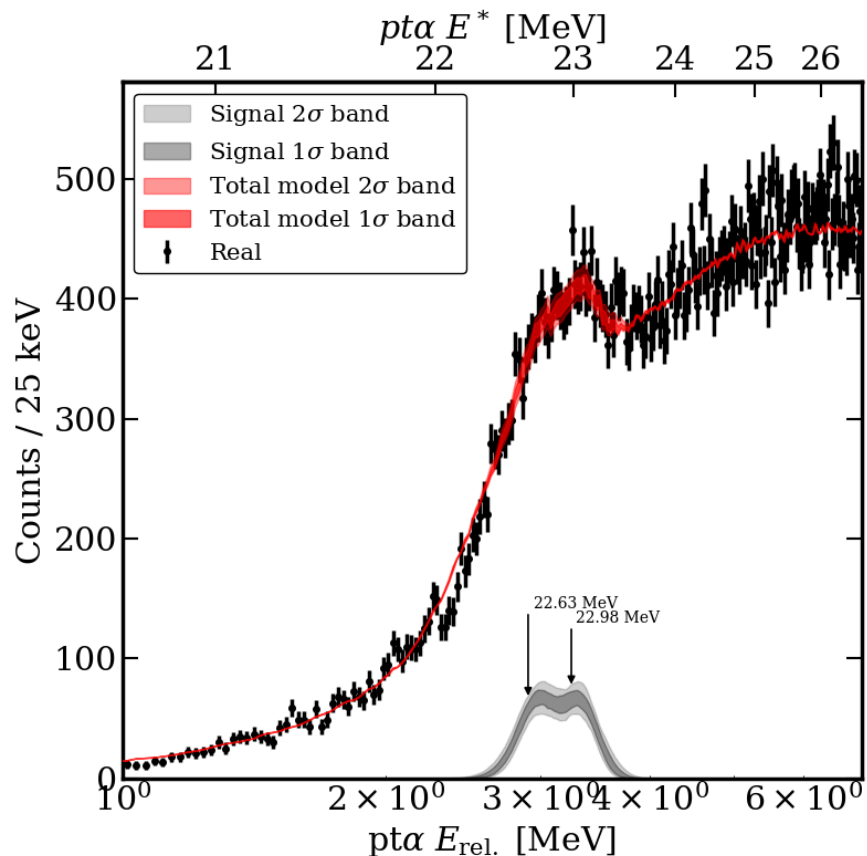


- Real resonances of ${}^8\text{Be}$ are present around $E^* = 23 \text{ MeV}$.
- 2-particle correlations are nonidentical
 - More partially mixed events

The data are visually inconsistent with a description of the data that excludes signals.

- A modified model which includes 3-particle correlations is required.

Another Case Study: $^8\text{Be} \rightarrow p + t + \alpha$



- Approximate double-Gaussian model leads to a reasonable total spectrum fit.

The data are visually consistent with a description of the data that includes a double-Gaussian signal.

- Real states are recovered within fitting error

Quick Recap

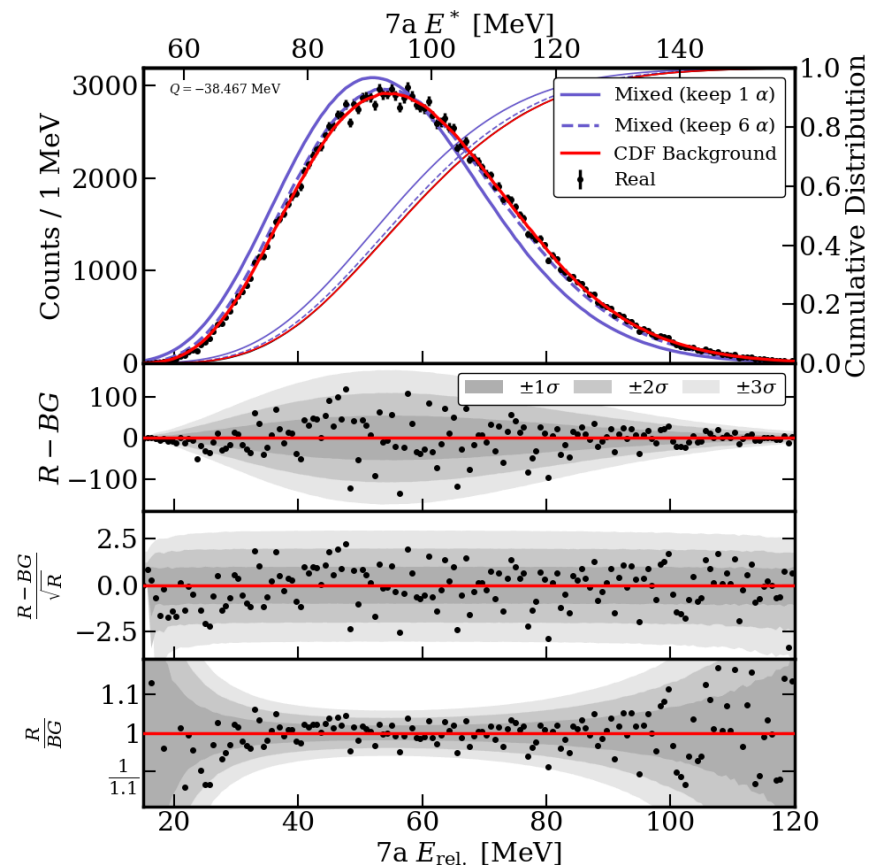
Mixed Events do NOT provide a reasonable description for the non-resonant background.

For systems with 3 or more particles, [Fully] Mixed Events can be corrected through the Partially Mixed Events.

In systems with physically no resonances, no signal needs to be added to describe the experimental data.

In systems with resonances, peaks must be incorporated into a total spectrum fit.

7α E^* spectrum (no signal model)



Number of α - α correlations

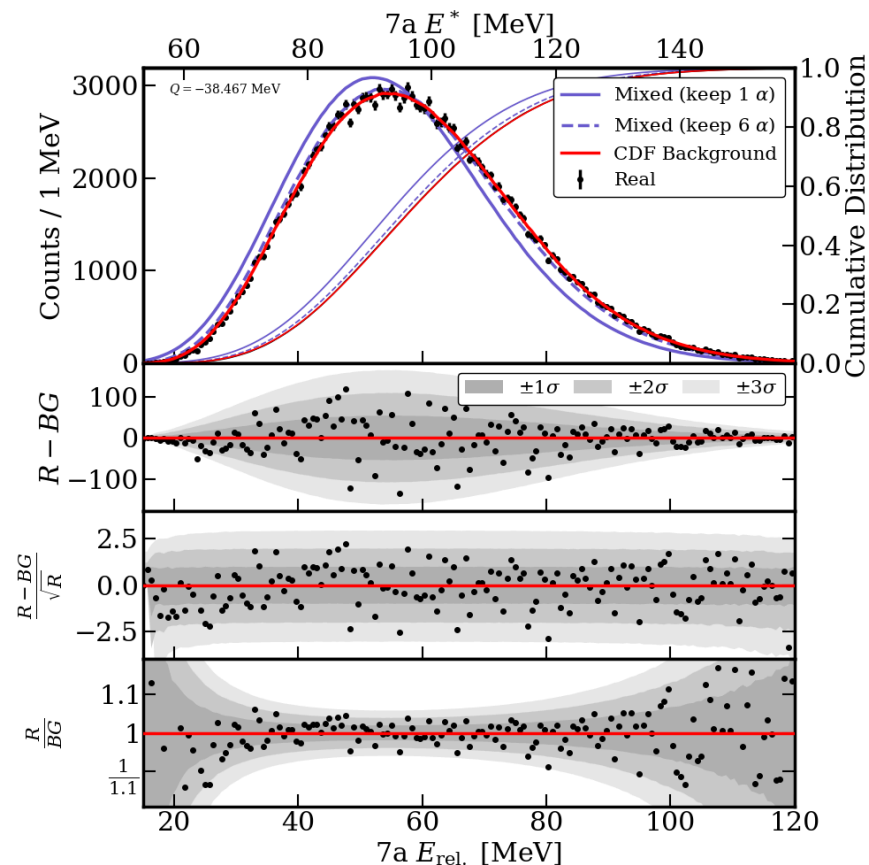
Real: $\binom{7}{2} = 21$

Partially Mixed: $\binom{6}{2} + \binom{1}{2} = 15$

Mixed: $7\binom{1}{2} = 0$

- Partial mixing incorporates 15/21 of the two-particle correlations

7α E^* spectrum (no signal model)



Number of α - α correlations

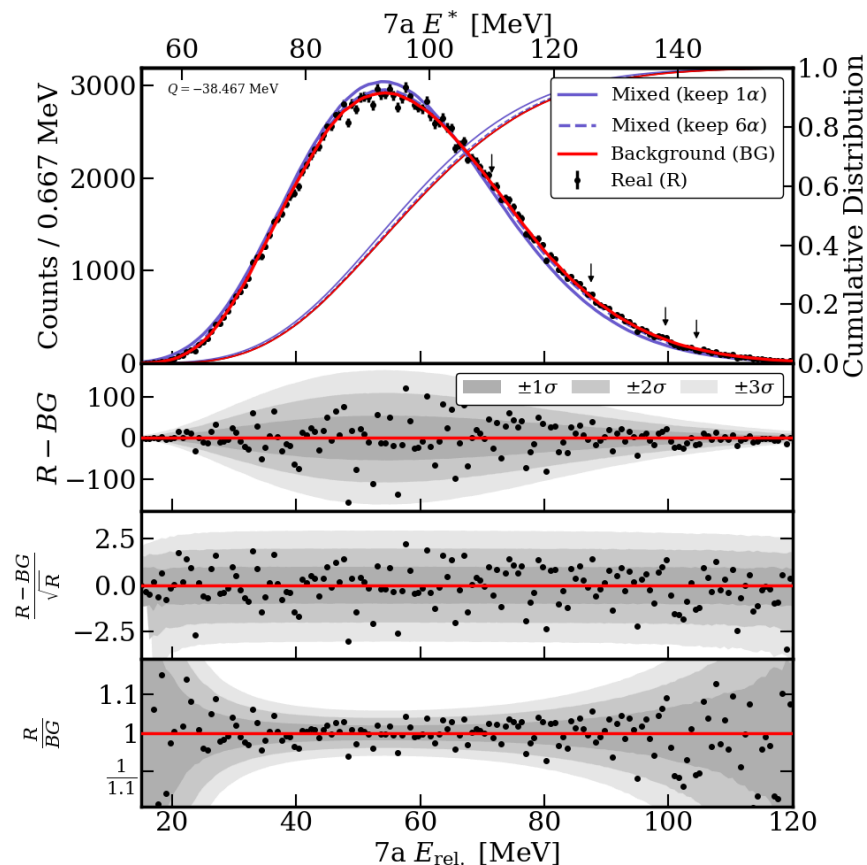
Real: $\binom{7}{2} = 21$

Partially Mixed: $\binom{6}{2} + \binom{1}{2} = 15$

Mixed: $7\binom{1}{2} = 0$

- Partial mixing incorporates 15/21 of the two-particle correlations
- Visually, pretty good description
- Minor deviation at low energy
 - Detector hit pattern bias

7α E^* spectrum (no signal model; hit pattern corrected)



- Modify mixing procedure so FM and PM result in same per-event hit pattern as the real data
- Great description of the data
 - $\chi^2 / \text{dof} = 1.09$ (1 is ideal)
 - $P = 0.52$ (0.5 is ideal)
- No perceivable systematic deviations
 - **Measured 7α distribution is statistically consistent with no resonances**
 - Further work needed to set upper limits of detection

Summary

Toroidal nuclei are exciting nuclear structures with decades of theoretical support.

Experimental investigations in 2018 motivated a series of higher-precision measurements of $N\alpha$ E^* spectra, searching for toroidal isomers in ^{28}Si , ^{32}S , and ^{36}Ar .

Deeper investigation of the biases of mixed events led to a much more accurate description of the background.

No statistically significant evidence for toroidal states was observed in the 7α E^* distribution of $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u.

- Upper limits studies are to follow (Likelihood ratio + MCMC).
- The $N\alpha$ systems of the ^{32}S and ^{36}Ar data are slated to be analyzed shortly.

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